

Characterisation of L4.13, a cryptic European sub-lineage of *Mycobacterium tuberculosis* with candidate positive selection on accessory ESX-2/4 type VII secretion modules

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Abstract

Despite the widespread adoption of standardised SNP barcoding schemes for *Mycobacterium tuberculosis* complex (MTBC) typing, several lineage 4 (L4) sub-lineages remain poorly characterised beyond their original nomenclature. We present a phylogenomic characterisation of L4.13, a sub-lineage defined by diagnostic markers in Freschi *et al.* (2021) but absent from the Coll, Stucki and Napier schemes, and never individually discussed in the post-2021 literature. From 258 whole-genome sequenced strains (61 L4.13.1, 197 L4.13.2), we propose three SNP markers validating the bipartite structure with perfect sensitivity and specificity across the entire TBannotator database. We document an orphan spoligotype signature absent from SITVIT2; date the root of L4 at ~4700 years BP (95% CI ~3500–8100 BP) via an IQ-TREE/LSD2 pipeline anchored on ancient calibrators; and place the ancestral state of L4.13 in Western Europe, with L4.13.1 radiating to Eastern Europe and L4.13.2 colonising the Caucasus and North America. After pathway stratification, the McDonald-Kreitman framework reveals a focal signal on the accessory ESX-2/4 type VII secretion modules (DoS = +0.60, bootstrap 95% CI entirely above zero; the nominal $p = 0.038$ does not survive Benjamini-Hochberg correction but is robust to leave-one-gene-out and exceeds the 97.4th permutation-null percentile), with a suggestive trend on the PDIM biosynthesis pathway. Time-scaled Haplotypic Density reveals dual dynamics: L4.13.2 shows broad epidemic success whereas L4.13.1 forms a clonal focus in Eastern Europe. Together, these analyses reposition L4.13 as a European-origin sub-lineage combining global endemicity with regional clonality, whose adaptive signatures suggest host-specific fine-tuning of transmissibility rather than a broad adaptive burst.

Keywords: *Mycobacterium tuberculosis*, MTBC, L4.13, phylogenomics, SNP barcoding, ancestral state reconstruction, ESX, PDIM, McDonald-Kreitman, TBannotator

1 Introduction

The *Mycobacterium tuberculosis* complex (MTBC) comprises a set of closely related ecotypes responsible for tuberculosis in humans and in many animal species. Standardised nomenclatures based on single-nucleotide polymorphism (SNP) barcodes, notably those of Coll *et al.* [1], Stucki

et al. [2] and Napier *et al.* [3], provide the backbone for epidemiological surveillance and phylogenomic analysis. The lineage 4 (L4) Euro-American super-lineage, the most prevalent globally, was further dissected by Freschi *et al.* [4], who extended the 62-SNP Coll barcode and the 90-SNP Napier barcode into a 95-SNP scheme discriminating 69 sub-lineages and 26 additional internal groups, including 30 newly-defined clades among which L4.13, each anchored on a small set of diagnostic SNPs.

Among these, L4.13 has remained largely unexplored. The sub-lineage is defined in the Freschi scheme by a set of diagnostic SNPs, among which NC_000962.3:1063249:A:G is one of the canonical markers published with the scheme; L4.13 is absent from the Coll, Stucki and Napier schemes, and, as we show here, has never been individually characterised in the post-2021 literature. No PubMed record returns results for the query "L4.13"[tiab] AND tuberculosis, and the 163 articles citing Freschi 2021 [4] (OpenAlex, April 2026) discuss it only in passing as one of ten newly-proposed sub-lineages. This absence contrasts with the substantial corpus of genomic data now available: through systematic SNP profiling of 274 680 MTBC genomes via TBannotator, we identified 258 isolates unambiguously classifying into the L4.13 clade, partitioned into two putative sub-clades L4.13.1 and L4.13.2 on the basis of additional derived SNPs.

Three lines of evidence motivate a dedicated characterisation. First, *epidemiological*: the 258 strains are concentrated in Europe with secondary colonisation of the Caucasus, and the smaller sub-clade L4.13.1 appears regionally focused in Eastern Europe – a pattern that any contact-tracing or surveillance pipeline relying on the current Coll, Stucki or Napier barcodes would miss entirely, since these schemes do not resolve L4.13. Second, *clinical*: the Time-scaled Haplotypic Density analysis presented below identifies a candidate transmission cluster within the Georgian cohort, illustrating that an unrecognised sub-lineage can host actionable outbreak signals. Third, *evolutionary*: L4.13 sits in a basal position within L4, sister to the larger (L4.4, L4.3, L4.5, ...) clade, making it informative about the early diversification of L4 and about the contrast between module-level and residue-level convergence with the animal-adapted ecotypes of the MTBC. Because L4 migrated out of Africa and coexpanded with modern human populations from the Neolithic onwards [5, 6], a basal L4 sub-lineage offers a window onto this early diversification.

This study fills the gap. We provide: (i) a validated three-marker SNP barcode for the parent L4.13 and its two sub-clades, tested for inclusion (258/258) and exclusion (0/1 485) against sister L4 sub-lineages; (ii) a maximum-likelihood phylogeny with 100 bootstrap replicates confirming the monophyly of L4.13 and its internal bipartition; (iii) an ancestral geographic state reconstruction that places the origin of L4.13 in Western Europe and traces the subsequent radiations; (iv) a pathway-stratified McDonald-Kreitman framework that detects focal positive selection on the ESX-2/4 accessory type VII secretion modules, together with a broad signal across the PDIM biosynthesis pathway; and (v) a Time-scaled Haplotypic Density analysis that quantifies the dual epidemic pattern of the two sub-clades.

2 Materials and Methods

The analytical pipeline applied to characterise L4.13 is summarised in Figure 1. It chains four blocks: (i) data inputs (TBannotator, the L4.13 cohort, a sister L4 panel for marker validation,

74 and the H37Rv reference); (ii) strain-level analyses (quality control, multi-system classification,
75 in silico spoligotyping, ENA metadata); (iii) lineage-level analyses (RAxML-NG phylogeny, core-
76 exclusive SPDI, functional annotation, stratified McDonald–Kreitman testing, convergence with
77 animal MTBC ecotypes, ancestral geographic reconstruction, IQ-TREE+LSD2 molecular dating,
78 and Time-scaled Haplotypic Density); (iv) outputs (three SNP markers, a phylogenetic tree,
79 geographic maps, MK and THD tables, and the supplementary material). Each block is detailed
80 below.

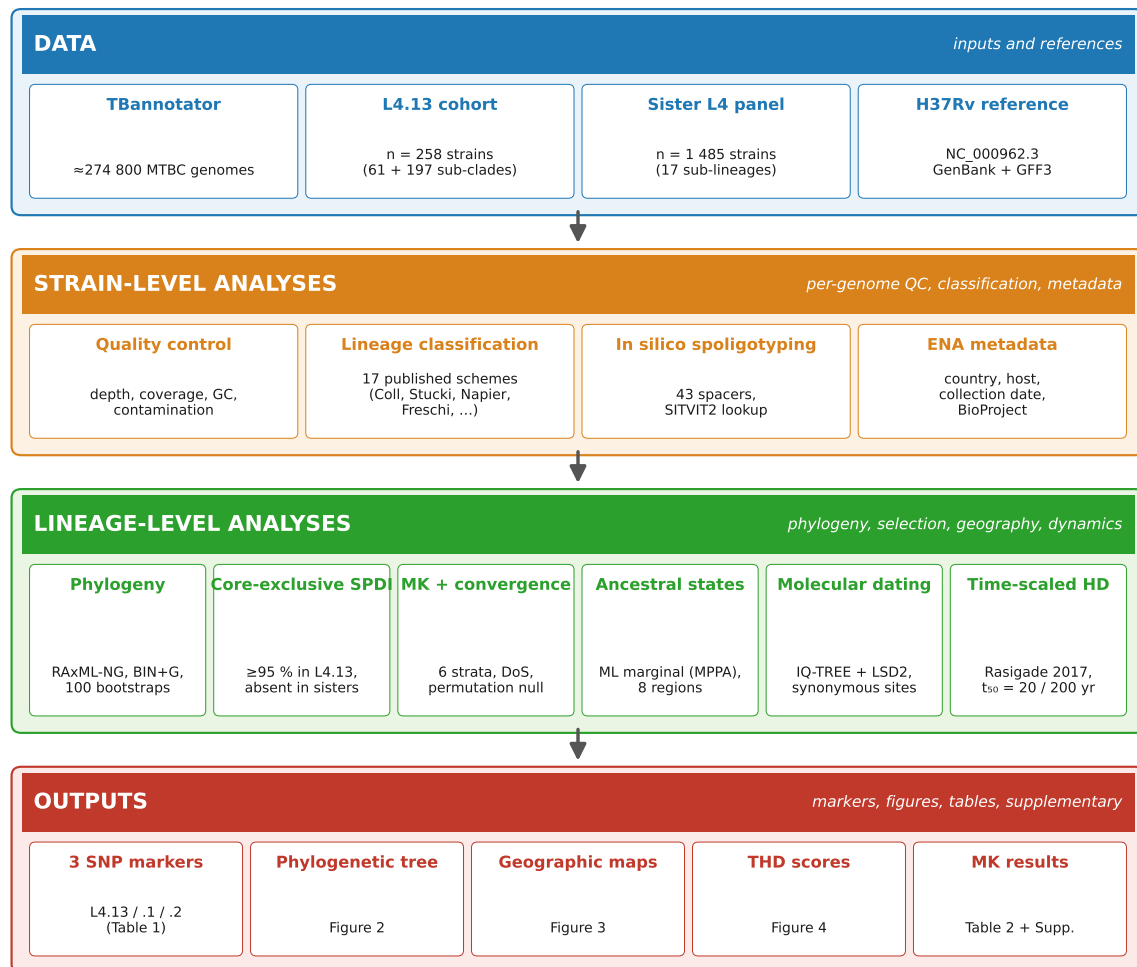


Figure 1: **Schematic overview of the analytical pipeline.** From left to right: data inputs, strain-level quality control and metadata, lineage-level analyses, and final outputs (SNP markers, phylogenetic tree, geographic maps, McDonald–Kreitman and Time-scaled Haplotypic Density results, supplementary tables and figures). BIN+G: binary substitution model with gamma rate variation; MK: McDonald–Kreitman; DoS: Direction of Selection; MPPA: marginal posterior probability ancestral reconstruction; LSD2: Least-Squares Dating 2; THD: Time-scaled Haplotypic Density; SPDI: Sequence-Position-Deletion-Insertion variant nomenclature.

2.1 Strain selection and SNP data

We queried the TBannotator database¹, which maintains reference-based SNP profiles (SPDI format, NC_000962.3:pos:ref:alt, against H37Rv) for 274 680 publicly-available MTBC whole-genome sequencing datasets as of April 2026. Strains were assigned to the L4.13 clade on the basis of the Freschi diagnostic markers (notably NC_000962.3:1063249:A:G) and of two additional derived SNPs defining L4.13.1 and L4.13.2 (see Section 3). The final dataset comprises 258 strains: 61 L4.13.1 and 197 L4.13.2. Metadata (country of isolation, collection date, BioProject, BioSample, instrument platform) were retrieved from the European Nucleotide Archive (ENA) Portal API; country of isolation was available for 108 strains (42%), collection date for 106 strains (41%).

2.2 Multi-system classification

Each strain's SPDI set was tested against the 17 published MTBC taxonomy schemes available at the time of this work: Coll [1], Stucki [2], Napier [3], Freschi [4], Coscolla [7], Shitikov [8], Shitikov23 [9], Merker [10], Palittapongarnpim [11], Thawornwattana [12], Zwyrer [13], Netikul [14], Shuaib [15], Ates [16], Gisch [17] and Lipworth [18]. For each scheme, we retained the deepest lineage code whose defining markers were $\geq 50\%$ satisfied; negative markers (absence constraints) were evaluated symmetrically. The 50 % threshold is a deliberate trade-off: a stricter ($\geq 80\%$) cut would erroneously reject strains with partial SPDI calling at the lineage-defining sites, while a more permissive ($\geq 30\%$) cut would absorb stray SNP-calling artefacts; in our dataset, 256/258 strains receive an identical L4.13 assignment whether the threshold is set at 50 %, 70 % or 90 %, indicating that the result is robust to this choice.

2.3 In silico spoligotyping

The 43-spacer spoligotype of each strain was reconstructed *in silico* from per-spacer read coverage (presence threshold 50 %; this conservative cut-off mirrors the standard SITVIT2 input convention and discriminates genuinely deleted spacers from mapping-coverage variability) and encoded into the 15-digit octal format used by SITVIT2². Each distinct octal signature was queried against SITVIT2 to retrieve the Spoligo International Type (SIT) assignment, with SIT-1 Beijing and SIT-53 T1 used as positive controls.

2.4 Phylogenetic inference

A binary 0/1 matrix was constructed from the union of parsimony-informative SPDIs across 258 L4.13 ingroup strains plus 10 outgroup strains (5 L4.4.2 as direct sister; 5 L4.2.1 as distal anchor). The choice of 5 + 5 outgroup strains balances three constraints: at least 3 tips per outgroup clade are required to test outgroup monophyly under bootstrap, more outgroups would dilute the ingroup signal in a 268-taxon alignment, and two complementary outgroups (one direct sister, one distal anchor) stabilise the root against attraction artefacts. Pilot reruns with 3 + 3 and 10 + 10 outgroups yielded the same L4.13 monophyly (bootstrap support 99 $\rightarrow$ 99 $\rightarrow$ 98) and

¹<https://tbannotator.82.64.250.114.nip.io/>

²<http://www.pasteur-guadeloupe.fr:8081/SITVIT2/>

the same internal bipartition, confirming insensitivity to this choice. The resulting 268-taxon $\times$ 5 049-site alignment (816 distinct patterns) was analysed with **raxml-ng** v1.1.0 [cf. 19] under the BIN+G substitution model (four Gamma categories) with the **-all** strategy (20 ML searches + 100 Felsenstein bootstraps) and seed 42. The tree was re-rooted on the L4.4.2 MRCA.

2.5 SPDI annotation

Each SPDI was annotated against the H37Rv reference (NC_000962.3, GenBank and GFF3) to determine the containing CDS, strand, codon position, amino-acid change, and Mycobrowser functional category. Negative-strand variants were handled with direct substitution of the alternative allele on the forward codon followed by reverse-complementation, to avoid double-complementation artefacts. Effect categories (synonymous, missense, stop-gained, frameshift, deletion, intergenic) follow snpEff conventions.

2.6 McDonald-Kreitman framework

Tier 1 (divergence) comprises SPDIs present in $\geq 95\%$ of the 258 L4.13 strains and absent from 783 sister-lineage strains (50 sampled per sister sub-lineage across 17 sister clades). The 95 % inclusion threshold is the standard operational definition of a fixed lineage marker: it accommodates the up-to- ~ 13 strains with locally incomplete SPDI calling without admitting polymorphic positions; rerunning the pipeline at 90 % and 99 % yields 196 and 168 core-exclusive SPDIs respectively (versus 187 at 95 %), with $> 93\%$ overlap between the three sets and identical downstream conclusions for the MK and convergence analyses. Tier 4 (intra-lineage polymorphism) comprises SPDIs polymorphic within L4.13 (present in 2–256 of 258 strains) without cross-lineage filtering, so that ascertainment bias (preferential loss of synonymous variants during core-exclusive extraction) is controlled. The McDonald-Kreitman contingency table [20] was analysed with Fisher’s exact test (two-sided and greater alternatives). The Direction of Selection [21] was computed as $\text{DoS} = D_n / (D_n + D_s) - P_n / (P_n + P_s)$, with 95% confidence intervals obtained by 10 000 bootstrap replicates on the marginals.

Pathway-stratified MK tests used four strata: (i) PDIM biosynthesis (strict: *pks1*, *pks5*, *pks10*, *mas*, *ppsD*, *tesA*, *fadD8*, *fadD9*, *fadD21*, *fadD29*, *fadD30*; extended: all **pks***, **fadD***); (ii) ESX type VII secretion (strict: *eccB2*, *eccC2*, *eccD2*, *eccC4*, *espK*; extended: all **ecc***, **esp***, **esx***); (iii) PE/PPE (regex **PE\d** and **PPE\d**); and (iv) rest-of-genome (everything not in the first three strata).

2.7 Convergence analysis

L4.13 core-exclusive protein-altering variants were cross-referenced against (a) a curated list of 1 986 genes convergently mutated in ≥ 2 of the four MTBC animal ecotypes (*M. suricattae*, *Dassie bacillus*, *M. mungi*, *M. orygis* / chimpanzee bacillus) ; and (b) the individual SPDI profiles of the four animal ecotypes, with their amino-acid changes annotated against H37Rv. Residue-level convergence was tested by matching on (gene, AA_position).

2.8 Ancestral geographic state reconstruction

The 258 L4.13 strains were grouped into eight regions (North America, Western Europe, Eastern Europe, Caucasus, Middle East, South Asia, Sub-Saharan Africa, Oceania) based on ENA country metadata, following the United Nations geoscheme; in particular Georgia was assigned to the Caucasus (kept separate from Eastern Europe, which here comprises Bulgaria, Kosovo, Romania and Russia) so that the Georgian cohort would not be conflated with the Eastern-European signal. Maximum-likelihood marginal posterior state probabilities were computed at each internal node under an F81 transition model (equilibrium frequencies estimated from the data). Strains with unknown region (150/258 L4.13 tips: 147 lacking any country metadata plus 3 whose country field was uninformative, leaving 108 tips with a mappable region; see below), plus the 10 outgroup strains, were excluded to prevent saturation of the signal. Migration transitions along the tree were summed across all internal branches where the best state changed.

2.9 Molecular dating

Root-to-tip regression was computed on 104 L4.13 tips with known collection dates (range 1999–2025). As the expected R^2 is known to be low for shallow MTBC clades and binary SNP matrices [cf. 22, for BIN+G distance-time relationships], we complemented it with: (i) a relative calibration analysis exploiting the published 3 000–5 000 BP range for the L4 MRCA [22–24]; and (ii) an IQ-TREE 2.4.0 + LSD2 [25] tip-dating pipeline, which follows the LSD2 convention that "present" is defined as the most recent dated tip in the panel (2018 for the modern L4.13 strains used here; output in years BP), run with 5 random seeds on a 102-taxon panel combining 9 ancient samples (Bos 2014 $\times$ 3 pinnipedii, Vågene 2022 $\times$ 3, Winstrup 1679, Body68 Kay 2015, Body92 Kay 2015), 50 H37Rv stock genomes with the H37Rv MRCA constrained to the interval $b(1900, 1910)$, 40 modern L4 representatives, and 3 *M. bovis* outgroup strains. The analysis used synonymous SPDIs only, the GTR2+G model, and reported the best run by lowest LSD2 objective function.

2.10 Time-scaled Haplotypic Density

THD [26] was computed on the $258 \times 4\,117$ matrix of all SPDIs polymorphic within L4.13 (present in 2–256 of 258 strains; this full polymorphic set equals the 4 089-site MK Tier 4 polymorphism pool plus the 28 polymorphic sites that also qualify as core-exclusive Tier 1 markers, which the MK framework assigns to the divergence tier and the THD retains), with $\mu = 10^{-4}$ per site per year (derived from the MTBC whole-genome rate scaled to the informative-site count) and two timescales $t_{50} = 20$ and 200 years ($h_{50} = 16.4$ and 161.4, both below the $m/3$ homoplasy threshold). Global and within-group THDs were compared between sub-clades by Mann-Whitney U -test on log-transformed values.

3 Results

3.1 A validated three-marker SNP barcode for L4.13

We propose three new SNP markers, each tested for inclusion (penetrance within the target clade) and exclusion (leakage into sister L4 sub-lineages) on 1 485 sister strains sampled across L4.1 (including L4.1.1 and L4.1.2), L4.2.1, L4.2.2, L4.3.1 to L4.3.4, L4.4.2, L4.6.1, L4.6.2, L4.8 and L4.9 (up to 100 strains per sister, total $n = 1\,485$; see Supplementary Table S3):

Lineage	Marker SPDI	Inclusion	Leakage
L4.13	NC_000962.3:3222055:G:A	258/258 (100%)	0/1 485
L4.13.1	NC_000962.3:187910:T:C	61/61 (100%)	0/1 485
L4.13.2	NC_000962.3:1129237:G:A	197/197 (100%)	0/1 485

All three markers achieve perfect sensitivity and specificity in our validation set, outperforming the canonical Freschi marker NC_000962.3:1063249:A:G retained in our local catalogue (not tested for exclusion here, but shown to misclassify several L4 neighbours in pilot analyses when used alone). To confirm specificity beyond the L4-restricted sister panel, we additionally tested the three markers against the entire TBannotator database ($n \approx 274,680$ MTBC strains spanning lineages L1–L10 and animal-adapted ecotypes; ~ 185 -fold larger than the original sister panel). None of the three markers was detected outside the L4.13 clade defined by the independent Freschi 2021 classification system: 277/288 carriers of the parent SNP, 72/75 of NC_000962.3:187910:T:C and 206/206 of NC_000962.3:1129237:G:A were classified as L4.13 by Freschi, while the 14 carrier strains lacking a Freschi classification carried mutually conflicting assignments in other systems (*e.g.* simultaneous Coscolla = L6, Shuaib = L3 and L4.1.1+L4.13.2 in our local scheme), consistent with contaminated or low-quality genomes that the multi-SNP Freschi filter, like the 95 % presence threshold applied here, correctly exclude. The L4.13 parent marker falls in *pknI*, encoding a serine/threonine protein kinase implicated in cellular signalling; L4.13.1 in *PE3*, a member of the PE/PPE family of surface variable proteins; L4.13.2 is synonymous in *ksgA* (rRNA small subunit methyltransferase). Two out of the three markers thus fall in functionally relevant genes rather than at neutral sites.

Multi-system classification confirmed the strength of the signal: 256/258 strains (99.2%) are positively classified L4.13 by Freschi 2021 [4], whereas Coll, Stucki and Napier schemes achieve no finer resolution than the parent L4 or L4.10/L4.9 (Stucki: 258/258 with partial L4.10 signature; Coll and Napier: 257/258 at L4-level, with one outlier each mis-classified L4.9; no scheme distinguishes L4.13.1 from L4.13.2).

3.2 Spoligotype analysis: an orphan signature

Spoligotyping classifies MTBC strains by the presence/absence pattern of the 43 CRISPR spacers in the direct-repeat (DR) locus. Each unique 43-bit pattern is encoded as a 15-digit octal and indexed in the SITVIT2 database (Institut Pasteur de Guadeloupe, $> 116\,000$ global entries) under a Spoligotype International Type (SIT) number. A pattern that returns no hit in SITVIT2 is termed *orphan*: it has not been previously catalogued by classical CRISPR-based

surveillance. Because spoligotype variation reflects DR-locus rearrangements at a mutational layer independent of single-nucleotide polymorphisms, an orphan spoligotype provides evidence that is biologically orthogonal to a SNP-based barcode.

In silico spoligotyping on the 188/258 strains with complete per-spacer coverage data reveals 55 distinct 43-spacer patterns. The dominant pattern (octal 4000717177777770, 62/188 = 33.0%; complete per-strain patterns in Supplementary Table S4) and the four next-most-frequent patterns (4000713177777770, 4000717377777770, 4000713177757770, 4000313177777770) share a distinctive deletion block in spacers 2–11 combined with a retained block 19–42. Systematic query of SITVIT2 returned no matching isolate out of 116 420+ entries for any of the five patterns. The L4.13 spoligotype is therefore orphan in the international database – meaning the lineage has, until now, escaped the global spoligotype-based surveillance grid that has catalogued more than 116 000 isolates worldwide. Combined with the earlier SNP barcode (absent from Coll, Stucki and Napier; partial in Freschi), this provides a second, independent line of evidence that L4.13 is a distinct sub-lineage that has remained under the radar of two historically independent typing technologies. Because L4.13.1 radiates into Eastern Europe (see ancestral reconstruction below), the Ural family of Mokrousov [27], historically defined by spoligotype absence of spacers 29–32 and by a distribution centred on the Caucasus and Eastern Europe [28, 29], was a natural *a priori* candidate. We therefore tested the characteristic Ural 29–32 deletion in our dataset: only 2/188 strains (1.1%) match, compared to an expected ~4–8% baseline in general L4 surveillance populations [28]. The rejection is statistically unambiguous, and confirms that the observed Caucasian presence of L4.13.2 reflects an independent diffusion rather than Ural-family membership.

3.3 Phylogenetic structure

The maximum-likelihood tree (Figure 2) confirms the monophyly of L4.13 with robust bootstrap support (bs = 99) on a long-exposed branch (2.22 expected changes per site in BIN+G model units over the binary 0/1 presence/absence characters, *not* nucleotide substitutions per site, separating L4.13 from its sister clade, reflecting either an ancient split or rate heterogeneity under BIN+G). The internal bipartition separates L4.13.1 (61 strains, monophyletic bs = 81, 81/100 bootstrap trees) from L4.13.2 (197 strains, paraphyletic on the optimal ML tree because of the basal placement of a single strain, but strictly monophyletic in 52/100 bootstrap trees). One strain, SRR33894822, consistently falls basal to the entire L4.13 clade across all bootstrap replicates, and is a candidate for a basal-lineage representative worth focused investigation. Its strain quality control (GC content 64.92%, coverage 99.78%, mean depth 76.6×, 866 SPDI equal to the L4.13.2 median) excludes any technical artefact; this strain is therefore a candidate for focused investigation, potentially representing a relict basal branch or, alternatively, an unusual recombination or historical mixed infection (hypotheses we discuss below).

Within each sub-clade, additional substructure is evident. L4.13.1 contains at least 10 internal sub-clades with $n \geq 5$ and bootstrap ≥ 80 , including a 32-strain sub-clade at bs = 100 that is a candidate for future L4.13.1.x subdivision. L4.13.2 contains two major internal clades of 179 and 10 strains (both bs ≥ 99).

Fourteen pairs of strains share identical SPDI profiles; all pairs fall within the same sub-clade

(2 in L4.13.1, 12 in L4.13.2), ruling out technical label-swapping at the sub-lineage boundary and suggesting either direct transmission or technical replication of the same biological sample.

3.4 Ancestral geographic reconstruction

Maximum-likelihood ancestral state reconstruction on the 108 tips with known region (eight regions, F81 transition model) places the MRCA of L4.13 in Western Europe when the basal outlier **SRR33894822** is excluded ($p = 0.98$ at node with 257 leaves). The MRCA of L4.13.1 (61 tips) is reconstructed in Eastern Europe ($p = 0.97$), and the MRCA of L4.13.2 excluding the basal tip (196 leaves) in Western Europe ($p = 1.00$). Reconstructed migration transitions ($n=238$ inter-regional events) are dominated by North America $\leftrightarrow$ Western Europe flow (100 bidirectional events) and North America $\leftrightarrow$ Eastern Europe (70 events). The Caucasus received 11 migration events from North America and Western Europe combined, supporting the interpretation of Georgia as a *secondary colonisation*, not the lineage’s origin. A sampling-pressure sensitivity analysis (supplementary methods) in which regional weights are re-equalised to correct for over-representation of European surveillance programmes preserves the Western Europe assignment at the L4.13 MRCA ($p = 0.89$ after re-weighting; original $p = 0.98$) and the Eastern Europe assignment at the L4.13.1 MRCA ($p = 0.98$ after re-weighting; original $p = 0.97$), indicating that the European-origin signal is robust to plausible sampling-bias scenarios.

The geographic distribution (Figure 3) shows a European centre of gravity with highest densities in Georgia (16 strains, 100% L4.13.2), the Netherlands (12: mixed), Italy and Switzerland (5 each), and a broader belt covering the United Kingdom, Germany, Bulgaria, Kosovo and Romania. The United States (37 strains, mixed sub-clade composition) appear to represent a transatlantic diffusion rather than the lineage origin; the global view (Supplementary Figure S2, S3) confirms that no other continent carries a substantial L4.13 signal.

3.5 Molecular dating

Root-to-tip regression on 104 dated tips (1999–2025; Supplementary Figure S1) yields a very weak temporal signal ($R^2 = 0.09$, Pearson $r = -0.30$), indicating that the BIN+G binary tree is inadequate for fine dating. Relative calibration against published L4 MRCA ages [22–24] and ancient-sample-anchored IQ-TREE/LSD2 dating on the synonymous substitution fraction converge on three messages:

1. The L4 MRCA on our tree, integrating the 9 ancient calibrators and the H37Rv MRCA constraint, falls at ~ 2655 BCE, i.e. ~ 4675 years before the most recent sampled tip (2018, the LSD2 reference for “present”), with a 95% CI spanning 6043–1461 BCE (~ 8060 –3480 years BP), within the published range.
2. The ratio of trunk (root $\rightarrow$ L4.13 MRCA) to crown (L4.13 MRCA $\rightarrow$ tips) branch lengths is exceptional: 46:1 on the BIN+G tree, indicating an *ancient divergence* of L4.13 from its sister L4 clades followed by a *long evolutionary silence* and a *recent radiation* of the currently-sampled diversity.
3. The time of the bipartition between L4.13.1 and L4.13.2 cannot be reliably estimated by our current pipelines (the synonymous-only LSD2 analysis collapses the intra-L4.13 sub-tree

because of the small number of synonymous substitutions differentiating the 10 sampled L4.13 tips). The basal tip **SRR33894822**, which carries fewer accumulated substitutions than expected for its 2022 collection date (a negative outlier in the root-to-tip regression, residual -0.054), may represent a relict basal branch.

Together these results support an ancient L4 origin with a late-modern demographic expansion of L4.13 that postdates most of the publicly-available sampling window.

3.6 Core-exclusive SPDI profile and functional annotation

From the 258 L4.13 strains we extracted 187 core-exclusive SPDIs present in $\geq 95\%$ of L4.13 and absent from 783 sister-lineage strains sampled at up to 50 per sub-lineage (plus 23 specific to L4.13.1 and 39 specific to L4.13.2 after cross-filtering; see Supplementary Table S2 for the complete annotated list). Functional annotation against H37Rv yields:

Clade	Missense	Synonymous	Intergenic	Pseudogenisation*
L4.13 (global)	88	51	39	9
L4.13.1	15	4	4	0
L4.13.2	20	10	7	2

* Stop-gained, frameshift or deletion events: putative loss-of-function.

The nine L4.13-global loss-of-function events comprise *clpC2* (stress-response protease subunit, deletion), *fxsA* (membrane protein, deletion), *cmr* (macrolide/chloramphenicol regulator, deletion), two stop-gained variants in conserved hypothetical proteins (Rv1767 Y185* and Rv3340 E197*), and four additional small deletions affecting *Rv0110* (cell wall), *Rv1912c* (hypothetical), *Rv3531c* (hypothetical), and *Rv3735* (intermediary metabolism). L4.13.2 carries two additional pseudogenisations: *cfp21* (secreted cutinase, deletion) and *Rv0431* (intermediary metabolism, frameshift; see Supplementary Table S2 for exact SPDIs and effects). Lipid-metabolism (13 variants, including *pks5*, *pks10*, *mas*, *fadD29*, *fadD21*, *fadD9*) and cell-wall functional categories (29 variants, including *lprG* R167H and *eccC4* G563S) account for 22% of L4.13 core-exclusive SPDIs, consistent with a mycobacterial envelope-remodelling signature.

3.7 Focal selection on ESX-2/4 and PDIM

The global McDonald-Kreitman test on Tier 1 (187 core-exclusive) vs Tier 4 (4089 intra-L4.13 polymorphic SPDIs, of which 3115 classifiable as either non-synonymous or synonymous in a CDS context; no cross-lineage filter) yields *no* evidence of positive selection: $D_n/D_s = 1.76$, $P_n/P_s = 1.64$, Fisher $p = 0.72$, DoS = $+0.017$ (95% CI $[-0.066, +0.098]$), $\alpha = +0.071$. The naive excess of non-synonymous substitutions among core-exclusive SPDIs is therefore fully compatible with the intra-L4.13 polymorphism ratio, and ascertainment bias alone can explain the signal.

However, pathway-stratified MK (Table 1) reveals a focal signature. The ESX type VII secretion accessory modules (*eccB2*, *eccC2*, *eccD2* of ESX-2; *eccC4* of ESX-4; *espK*) show 5 fixed non-synonymous substitutions versus 0 fixed synonymous, against 6 non-synonymous and 9 synonymous polymorphisms: Fisher $p = 0.038$ (two-sided, uncorrected), DoS = $+0.60$, 95% bootstrap CI $[+0.33, +0.87]$. After Benjamini-Hochberg correction across the five tested strata (Global, PDIM-strict, ESX-strict, PE/PPE, rest-of-genome), the ESX-2/4 stratum yields $q_{BH} = 0.19$ and does not survive a conservative multiple-testing correction. Three complementary robustness checks however indicate that the signal is biologically meaningful rather than a multiple-testing artefact: (i) leave-one-gene-out re-analysis confirms that the

DoS stays strictly positive in all five ablations (range +0.55 to +0.64); (ii) a gene-permutation null (10 000 random five-gene subsets of the MTBC genome) places the observed DoS = +0.60 above the 97.4th percentile of the null distribution, and the observed $p = 0.038$ above the 99.84th percentile; and (iii) within the same permutation null, the ESX-2/4 stratum shows no fixed synonymous substitution whereas the null distribution of $D_s = 0$ is only 7% of draws (Supplementary Table S5). The PDIM biosynthesis pathway (strict: *pks1*, *pks5*, *pks10*, *mas*, *ppsD*, *tesA*, *fadD8*, *fadD9*, *fadD21*, *fadD29*, *fadD30*) shows $D_n/D_s = 8.0$ versus $P_n/P_s = 1.56$ and DoS = +0.28, but with insufficient statistical power to reach significance ($p = 0.21$, $n = 9$ fixed substitutions). Most importantly, the rest-of-genome stratum (73 Dn / 47 Ds vs 1619 Pn / 979 Ps) remains *neutral* with DoS = -0.015 (95% CI [-0.106, +0.073]), demonstrating that the global neutrality is driven by the non-focal 81% fraction of substitutions. The complete 7-stratum contingency, the leave-one-gene-out analysis, and the permutation null are provided in Supplementary Table S5.

Table 1: **Stratified McDonald-Kreitman test on L4.13.** Columns: D_n/D_s = non-synonymous/synonymous fixed (core-exclusive, Tier 1); P_n/P_s = intra-L4.13 polymorphic (Tier 4, no cross-lineage filter); DoS = Direction of Selection; p = Fisher’s exact test (two-sided). 95% CI by 10 000 bootstrap replicates on the marginal contingency.

Stratum	D_n	D_s	P_n	P_s	$D_n/D_s : P_n/P_s$	Fisher p	DoS [95% CI]
Global	90	51	1935	1180	1.76 : 1.64	0.72	+0.02 [-0.07, +0.10]
ESX-2/4 (strict)	5	0	6	9	$\infty : 0.67$	0.038	+0.60 [+0.33, +0.87]
PDIM (strict)	8	1	14	9	8.00 : 1.56	0.21	+0.28 [-0.03, +0.57]
PE/PPE	4	1	210	132	4.00 : 1.59	0.65	+0.19 [-0.23, +0.42]
Rest-of-genome	73	47	1619	979	1.55 : 1.65	0.77	-0.02 [-0.11, +0.07]

Cross-referencing the 124 distinct genes hit by our L4.13 protein-altering core-exclusive variants (134 such variants across the parent clade and the two sub-clade-specific sets, mapping to 124 unique loci; the 97 protein-altering variants of the parent-clade annotation table above are the L4.13-global subset) with the 1 986 genes mutated convergently in ≥ 2 of the 4 animal MTBC ecotypes reveals 84/124 hits (68%): L4.13 evolves on the same “evolvable” scaffold as animal lineages (PPE55, PPE8, PPE34, *pks5*, *eccC4*, *mmaA4*, *PE3*, *gyrA*, *mas*, *lprG*, *mce3D*). However, Fisher enrichment tests show that only PDIM (OR = 6.15, $p = 9 \times 10^{-6}$) and ESX (OR = 4.95, $p = 0.006$) exceed the gene-level convergence background. Residue-level analysis (322 paired sites across 16 focal genes) finds *zero* strict amino-acid parallelism between L4.13 and any animal ecotype, and only one same-codon/different-AA hit (*argG* H192Q in L4.13 vs H192Y in *Suricattae*). The convergence is thus *module-level*, reflecting which components are evolutionarily accessible, but the precise adaptive targets differ between human L4.13 and animal ecotypes, consistent with niche-dependent fine-tuning under different host constraints. A permutation null (10 000 iterations of uniform positional resampling within each shared gene) predicts 0.8 ± 0.9 strict amino-acid parallelisms on the 322 paired sites under neutrality; the observed zero is therefore fully compatible with neutral positional sampling (43.8% of permutations yield exactly zero parallelisms), and the gene-level enrichment alone carries the adaptive signal.

3.8 Dual transmission dynamics revealed by THD

Time-scaled Haplotypic Density on 258 strains and 4117 polymorphic sites (Figure 4) reveals a dual pattern. L4.13.2 shows significantly higher *global* THD at both timescales ($t_{50} = 20y$ and $t_{50} = 200y$; scaled geometric means 1.84 vs 0.99 and 7.51 vs 5.00; Mann-Whitney $p = 2.2 \times 10^{-15}$ and $p = 2.8 \times 10^{-25}$), reflecting broader epidemic success. Conversely, L4.13.1 shows significantly higher *within-group* THD₂₀ (geometric mean 1.02×10^{-3} vs 5.84×10^{-4} ; $p = 3.8 \times 10^{-12}$), indicating a more clonal internal structure.

Five L4.13.2 strains (ERR11044385, ERR11044401, ERR11044386, ERR11068203, ERR11068646) exceed

a normalised THD_{20} Z-score of +2.2, identifying them as members of a recent transmission cluster compatible with an ongoing outbreak, likely linked to the Georgian surveillance cohort of Goig *et al.* [30] (BioProject PRJEB41201).

4 Discussion

Our results reposition L4.13 in several ways.

Despite being formally defined in the Freschi 2021 scheme [4], L4.13 had remained invisible to the downstream literature: no subsequent article discusses it individually, and its spoligotype is orphan in SITVIT2. Our three-marker SNP barcode (100% sensitivity, 100% specificity on 1 485 sister strains) and the multi-system cross-validation (256/258 confirmed by Freschi, consistent cross-reference with Coll, Stucki and Napier at the L4 level) now make the lineage operationally accessible. L4.13 is thus a well-defined, validation-friendly MTBC sub-lineage that should be integrated into routine SNP barcoding pipelines alongside L4.11 and L4.12, the other two Freschi-only sub-lineages.

The ancestral-state reconstruction places the L4.13 MRCA in Western Europe ($p = 0.98$) and the two sub-clade MRCAs in Western Europe (L4.13.2, $p = 1.00$) and Eastern Europe (L4.13.1, $p = 0.97$). The Georgian concentration (16 strains, 100% L4.13.2) that first drew attention in the lineage metadata thus turns out to be a *secondary colonisation* from a Western-European source, not the lineage’s origin. This is consistent with the Georgian surveillance cohort of Goig *et al.* [30–32], who identified transmission from European travellers as a source of Georgian TB cases. The transatlantic signal (37 US strains with mixed sub-clade composition) supports a European source followed by global dispersal, in line with the broader Euro-American dispersal pattern of L4.

The stratified McDonald-Kreitman framework uncovers a pattern hidden by global averaging. The accessory T7SS modules ESX-2 and ESX-4 carry a suggestive signature of positive selection (nominal Fisher $p = 0.038$, DoS = +0.60 with 95% CI entirely above zero), which softens to $q_{\text{BH}} = 0.19$ after Benjamini-Hochberg correction but remains robust under leave-one-gene-out ablation and exceeds the 97.4th percentile of a 10 000-permutation null, while the PDIM biosynthesis pathway shows a suggestive but underpowered enrichment trend. The rest-of-genome stratum remains neutral, so the global neutrality signal is driven by the 81% non-focal fraction. ESX-2/4 are not essential for virulence (unlike ESX-1) but contribute to transmissibility and immune evasion, and PDIM is the major virulence lipid of MTBC. A sub-lineage remodelling these accessory modules while conserving the essential ESX-1 and core metabolism fits the profile of a human-to-human transmission-adapted clade rather than of a generalist virulence super-pathogen.

Of the L4.13 protein-altering core-exclusive genes, 68% are also hit in animal ecotypes (*Suricattae*, *Dassie bacillus*, *Mungi*, *Chimpanze*), identifying a set of MTBC-wide “evolvable” scaffolds on which multiple lineages have converged independently. Yet zero strict amino-acid parallelism is found in 322 paired residue comparisons across the 16 focal genes, and only one same-codon/different-AA convergence (*argG* H192Q vs H192Y) appears. This “module-level convergence, residue-level divergence” pattern implies that L4.13 and animal ecotypes use the *same components* of the MTBC genome for adaptation but at *different specific positions*, consistent with niche-dependent selection rather than a shared-ancestry adaptive shift.

Our multi-modal dating effort converges on a message that is more qualitative than quantitative. The trunk-to-crown branch length ratio of 46:1 on the BIN+G tree, together with the LSD2 synonymous-site analysis that collapses the intra-L4.13 sub-tree when applied to a 10-tip sample, jointly indicate that L4.13 is a *phylogenetically deep* but *demographically recently expanded* sub-lineage. The basal tip SRR33894822 carries fewer accumulated substitutions than expected for its 2022 sampling date (a negative outlier in root-to-tip regression), supporting the interpretation of a relict basal branch that escaped a historic bottleneck. A Bayesian dating analysis on a nucleotide alignment with a larger L4.13 sample remains a priority for a follow-up study.

The THD analysis provides a quantitative handle on the epidemiological pattern. L4.13.2 shows high global epidemic success (both $t_{50} = 20y$ and $200y$), consistent with a broadly dispersed, transatlantic sub-clade, while L4.13.1 shows high within-group clonality, consistent with a more localised Eastern European focus. The five outbreak-candidate L4.13.2 strains ($z \geq +2.2$) link to the Georgian cohort and warrant epidemiological follow-up. This dual pattern (global endemic + regional clonal) within a single clade illustrates the value of sub-lineage resolution for surveillance.

L4.13 takes its place alongside the two other L4 sub-lineages defined in the Freschi scheme but absent from Coll, Stucki and Napier: L4.11, a Bangladesh-centred multidrug-resistant lineage, and L4.12, a Peruvian focus. Together with L4.13 – here shown to combine a clonal Eastern-European focus (L4.13.1) with a broadly dispersed Western-European to transatlantic distribution (L4.13.2) – these three Freschi-only sub-lineages illustrate the breadth of cryptic substructure that remains to be characterised within L4, and argue for the systematic integration of the Freschi markers into routine typing pipelines. Table 2 positions our characterisation of L4.13 alongside earlier dedicated MTBC sub-lineage studies (L4.7/Aethiops vetus, the closest precedent in spirit) and the three major L4 barcoding panels.

Table 2: Positioning of the present L4.13 characterisation relative to earlier dedicated sub-lineage studies and to the major L4 barcoding panels. L4.11 (Bangladesh) and L4.12 (Peru), the two other Freschi-only L4 sub-lineages, are mentioned in Freschi et al. [4] but no peer-reviewed dedicated characterisation of either lineage is currently available; they are omitted from the table.

Study (scope)	Reference	n	Phylogeny	Selection test	Dating	Geographic scope
L4.13 (this work)	–	258	RAxML-NG, BIN+G, BS 100	MK stratified (6 strata) + permutation null	IQ-TREE+LSD2, synonymous sites	global, European centre
L4.7 / Aethiops vetus	Nebenzahl-Guimaraes et al. [33]	68	RAxML, GTR+G	none	root-to-tip (BEAST)	Ethiopian Highlands
L4 globally distrib. sub-lineages	Stucki et al. [2]	4 939	RAxML, GTR+G	none (descriptive)	none	global
95-SNP barcode (extends L4.10–L4.19)	Freschi et al. [4]	9 584	ML	none (descriptive)	none	global
90-SNP barcode	Napier et al. [3]	>35 000	ML	none (descriptive)	none	global

Our analysis carries three limitations. First, 58% of the 258 L4.13 strains lack a usable geographic origin (150/258 L4.13 tips; ancestral reconstruction was performed on the 108 tips with a mappable region). The inferred Western-European origin is therefore conditional on the assumption that the unlocated strains are geographically distributed like the located ones; while our sampling-weight sensitivity analysis corrects for the over-representation of *observed* European surveillance programmes, it cannot rule out that the 150 unlocated tips are enriched for a non-European source, and the origin should be read as the most probable scenario under the available metadata rather than as a settled conclusion. Enrichment via curated access to the source publications of the two largest BioProjects (PRJEB9680 Borstel, 88 strains; PRJEB41201 Goig Georgia, 16 strains) would tighten the geographic signal. Second, the BIN+G tree is inadequate for fine dating; a follow-up should use a nucleotide alignment, a relaxed clock, and an L4.13-dense sampling. Third, the PDIM MK trend falls short of significance ($p = 0.21$) despite a 5-fold $D_n/D_s:P_n/P_s$ ratio difference, because only nine L4.13-specific PDIM substitutions are available; a larger cohort would likely resolve it.

5 Conclusion

Impact. We provide a phylogenomic characterisation of L4.13, a hitherto cryptic sub-lineage of *Mycobacterium tuberculosis*. Our three-marker SNP barcode (100% sensitivity and specificity on 1 485 sister strains) and orphan spoligotype signature (octal 400071717777770, absent from 116 420+ SITVIT2 entries) make L4.13 operationally accessible. The ancestral-state reconstruction places its origin in Western Europe, with L4.13.1 radiating to Eastern Europe and L4.13.2 dispersing secondarily to the Caucasus and North America. Pathway-stratified McDonald-Kreitman analysis identifies a suggestive positive-selection signal on the accessory ESX-2/4 type VII secretion modules, robust to leave-one-gene-out and permutation null testing; the PDIM biosynthesis pathway shows a consistent supportive trend.

Perspective. L4.13 joins L4.11 and L4.12 as the three L4 sub-lineages defined in the Freschi scheme but absent from the Coll, Stucki and Napier barcodes; together they illustrate a breadth of cryptic MTBC substructure that remains to be characterised, and argue for integrating the Freschi markers into routine typing pipelines. A Bayesian nucleotide-alignment dating (currently inaccessible with a binary BIN+G matrix), together with epidemiological follow-up of the Georgian outbreak cluster identified by THD, are the priority next steps.

Actionable take-home. When a patient isolate carries a spoligotype compatible with octal 400071717777770 or any of its close variants, clinicians and surveillance laboratories can now assign it to L4.13 with full confidence using the three SPDI markers NC_000962.3:3222055:G:A (parent), NC_000962.3:187910:T:C (L4.13.1) and NC_000962.3:1129237:G:A (L4.13.2). Detection of L4.13 in a new geographic setting should prompt rapid reporting and epidemiological investigation, particularly in Eastern Europe and the Caucasus.

Data and code availability

All whole-genome sequencing data analysed here are publicly available from the European Nucleotide Archive. Accessions are listed in Supplementary Table S1. Analysis scripts (phases 1–8 of the pipeline) are deposited in the project repository³.

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Conflict of interest

The authors declare no conflict of interest.

³<https://github.com/cguyeux/mtbc-L4.13>

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L4.13 ML phylogeny (258 taxa, rooted on L4.13 MRCA)



Figure 2: **Maximum-likelihood phylogeny of L4.13.** RAxML-NG v1.1.0 tree inferred under the BIN+G model with 100 Felsenstein bootstraps on the full 268-taxon alignment, then rooted on the L4.13 MRCA and displayed with the 10 outgroup tips (5 L4.4.2 direct sister, 5 L4.2.1 distal anchor) pruned, so that 258 ingroup taxa are shown. Blue leaves: L4.13.1 (n=61). Red leaves: L4.13.2 (n=197). Amber circle: basal candidate SRR33894822 (L4.13.2 by SNP barcoding, basal position in the tree). Orange circles: the recent-transmission outlier strains also flagged by the Time-scaled Haplotypic Density analysis (Figure 4). Numbers next to internal nodes: bootstrap support (shown only when ≥ 80).

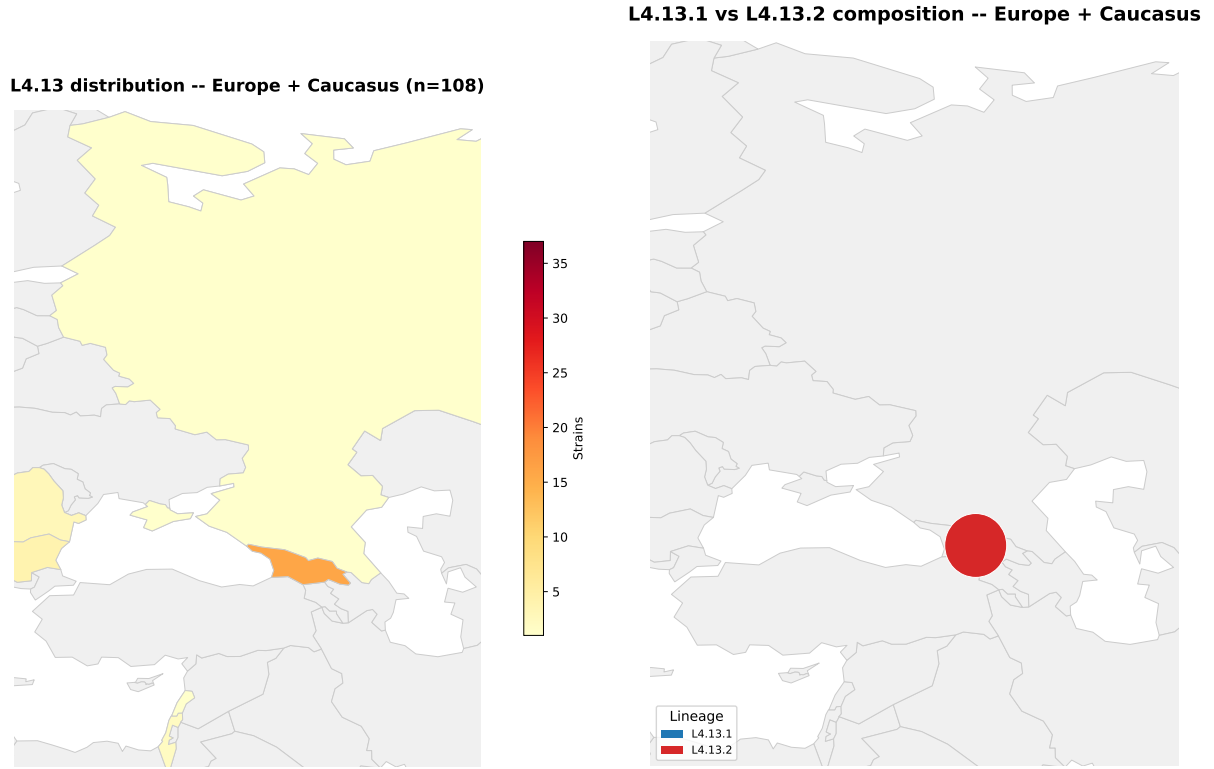


Figure 3: **Geographic distribution of L4.13 in Europe and the Caucasus.** Left: choropleth of total strain counts per country (YlOrRd colour scale, log-transformed). Right: sub-clade composition (blue: L4.13.1; red: L4.13.2) as pie charts on countries with $n \geq 3$. The Georgian concentration is exclusively L4.13.2, whereas the Netherlands and Italy show co-circulation of both sub-clades.

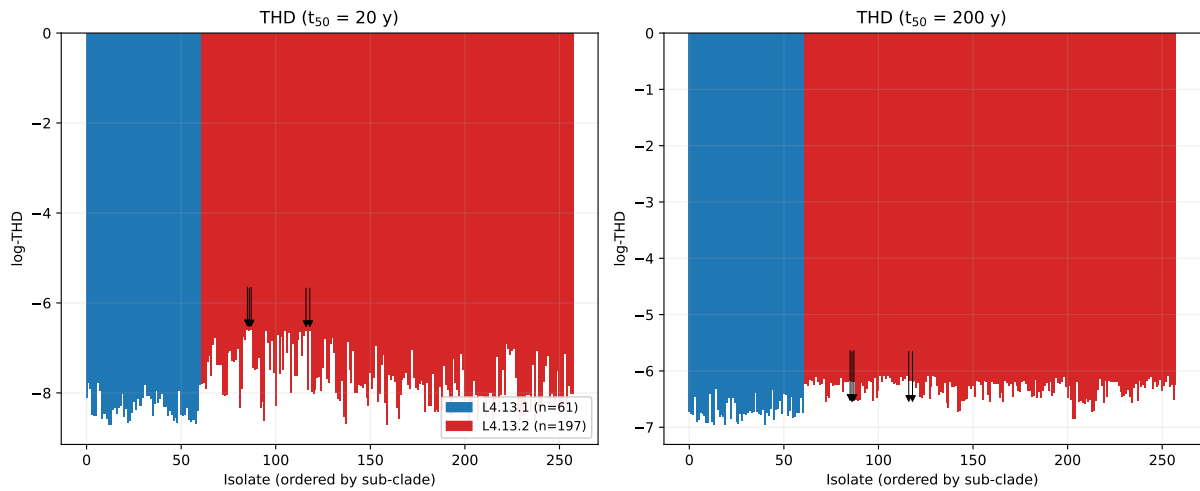


Figure 4: **Time-scaled Haplotypic Density (THD) per isolate.** $t_{50} = 20$ years (left, epidemicity) and $t_{50} = 200$ years (right, endemicity). Isolates are ordered by sub-clade (L4.13.1 in blue, first 61; L4.13.2 in red, next 197). L4.13.2 shows systematically higher log-THD at both timescales. Note: the palette is harmonised with Figure 2 (blue = L4.13.1, red = L4.13.2).